Optimization Dynamics Imprint Semantic Specificity in Contrastive Embedding Norms

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Abstract

Contrastive embedding models trained with scale-invariant losses are typically paired with distance metrics like cosine similarity, effectively ignoring embedding magnitudes. However, surprisingly, empirical studies reveal that despite this, these "discarded" norms seem to correlate with semantic properties such as concept specificity, token frequency, and human uncertainty. In this work, we provide a formal theoretical framework explaining this phenomenon. By analyzing the optimization dynamics, we derive an analytic formula demonstrating that embedding length naturally encodes this information as a byproduct of the training process. We also show how this gives rise to signals that can serve as "free" calibration tools in specific models and retrieval tasks, providing a grounded explanation for a previously heuristic observation.

1 Introduction

Contrastive embedding models now span visual self-supervision (SimCLR, MoCo [3, 7]), image-text pretraining (CLIP [21]), and text retrieval (Sentence-BERT, E5 [22, 28]). Their objectives match positives against negatives with dot products or cosine similarities after mapping each embedding z to $z/\|z\|$, making $\|z\|$ irrelevant to every training logit and to standard cosine inference. This design choice stabilizes logit scale, makes temperature meaningful, encourages hyperspherical geometry, and removes magnitude from the stated objective.

Yet embedding norm is repeatedly reported as a signal. Prior work links norm to confidence-like quantities in self-supervised learning [6], sample quality in face recognition [16, 10], information gain in word embeddings [18], and uncertainty in probabilistic contrastive learning [11, 12]. Retrieval results sharpen the puzzle: cosine-only scoring underestimates similarity for high-frequency words, and explicit norm discounting partly repairs the error [41, 33]. Existing evidence therefore creates a tension: the loss is designed so $\|z\|$ cannot affect the optimized similarity, but trained models still encode useful information in $\|z\|$. What remains missing is a mechanism that predicts when norm grows, when it shrinks, and which direction a norm-aware correction should take.

The main contribution of this paper is to provide this missing mechanism. We show that the norm of a fixed input is governed by its gradient update uncertainty and drift, which can be intuitively interpreted as the specificity of the concept the input represents and how awkwardly it fits between other inputs. Central to this theory is an analytic equation predicting quantitatively how this happens, explaining previously puzzling effects. Norm thus becomes a free inference-time signal: when pure cosine similarity is suboptimal, inverse-norm discounting can improve zero-shot classification and retrieval without retraining or extra supervision; when cosine similarity is nearly optimal, norm still provides information about answer quality and uncertainty that traditional benchmarks usually omit.

Preprint.

2 Related Work

Work on hyperspherical representations explains why normalized contrastive objectives privilege directions. The standard reference point is the alignment–uniformity view of contrastive learning [31], under which normalized objectives encourage models to shape angular structure on the unit sphere. Hyperspherical metric-learning papers such as NormFace [27], Heated-Up Softmax Embedding [40], and Deep Metric Learning with Spherical Embedding [38] also note that normalization induces inverse-norm gradients and nontrivial radial dynamics, which helps explain why the design works. This at the same time suggests that norm cannot simply be ignored, though predicting and interpreting it is difficult.

Norm as an information carrier. Several papers show that norm carries information even when the loss removes it from similarity scoring. Draganov et al. [6] prove that scale-invariant losses produce tangential gradients that make weight norms grow monotonically and correlate with confidence-like quantities. MagFace [16] and AdaFace [10] use feature norm as a sample-quality proxy; Oyama et al. [18] connect word-embedding norm to information gain; Kirchhof et al. [11, 12] give probabilistic interpretations through von Mises–Fisher concentration and Monte Carlo InfoNCE. These works motivate radial information but leave open the mechanism we study: how specificity, uncertainty, and transmitted gradient drift jointly determine the equilibrium norm of an individual embedding.

Norm-aware similarity and geometry of models like CLIP. Retrieval work shows where radial information changes rankings. Cosine-only matching understates similarity for high-frequency words, and norm discounting partly repairs the error [41, 33]; cross-modal retrieval normalizations improve performance by suppressing hub-like gallery items [1]. In multimodal models, prior work documents modality gap, anisotropy, and nontrivial global geometry in CLIP despite normalized contrastive training [15, 26, 14]. We connect these phenomena to a stationary norm law: large norms can reflect uncertainty or hub structure rather than mere frequency, and the resulting correction should discount large norms rather than amplify them.

3 Setup and Notation

The theory tracks a contrastive embedding model which maps an input x (a sentence in a text encoder, an image or caption in a vision-language model) to an embedding vector $z = f(x;\theta)$, which can be used to compare similarity between inputs.

Inputs and concepts. Throughout, x denotes such an input, and θ_t denotes the parameter of the model after t optimizer steps. We study how the embedding $z_t(x) := f(x;\theta_t)$ changes as the parameters θ_t are updated during training.

Architecture. Typically, both the internal transformer layers and the final hidden state before the projection head undergo normalization (such as LayerNorm or RMSNorm), which re-centers and rescales representations to largely eliminate the effect of parameter magnitude within those stages. Consequently, the parameters *after* this final normalization, namely the projection-head weights, exert the most direct control over the ℓ_2 norm of the output embedding, while the rest of the network mainly governs direction rather than scale. For models without a projection head, the final layer weights play this role.

Training objective. The canonical example is the symmetric InfoNCE objective used in contrastive multimodal models. Given a batch of N matched pairs (x_i, y_i) from two modalities, let $z_i^{(x)}$ and $z_j^{(y)}$ denote the unnormalized embeddings from the respective encoders, and $u_i = z_i^{(x)}/\|z_i^{(x)}\|, \quad v_j = z_j^{(y)}/\|z_j^{(y)}\|$. The similarity matrix is computed as $S_{ij} = u_i^\top v_j/\tau$, where $\tau > 0$ is a learned temperature parameter. The contrastive loss for the x-to-y and y to x directions are:

$$\mathcal{L}_{x\to y} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(S_{ii})}{\sum_{j=1}^{N}\exp(S_{ij})}, \mathcal{L}_{y\to x} = -\frac{1}{N}\sum_{j=1}^{N}\log\frac{\exp(S_{jj})}{\sum_{i=1}^{N}\exp(S_{ij})}. \tag{1}$$

The total loss is $\mathcal{L} = (\mathcal{L}_{x\to y} + \mathcal{L}_{y\to x})/2$. This loss depends on embeddings only through direction, not magnitude. Our theory applies to such scale-invariant loss.

Assumption on the loss. InfoNCE is just a representative instance; throughout the paper we require only the following structural property of $\mathcal{L}$ (the specific form plays no role in the theory):

> **(A)** *Scale invariance.* For every i and every $\alpha > 0$,
> $$\mathcal{L}(z_1, \dots, \alpha z_i, \dots, z_M) = \mathcal{L}(z_1, \dots, z_i, \dots, z_M).$$

Assumption (A) holds whenever $\mathcal{L}$ computes similarities exclusively via ℓ_2-normalized embeddings, as in InfoNCE and its variants.

> **(B)** *Per-sample decomposition.* Given a minibatch $B_t = \{(x_k, y_k)\}_{k=1}^N$ of i.i.d. draws from P_{data}, the loss decomposes as
> $$\mathcal{L} = \frac{1}{N}\sum_{k=1}^{N} \mathcal{L}_k,$$
> where each $\mathcal{L}_k$ depends on the pair (x_k, y_k) and the other batch members only through the shared similarity scores.

Assumption (B) lets us write the parameter gradient as a sum of per-sample contributions, $\nabla_\theta \mathcal{L} = \frac{1}{N}\sum_k \nabla_\theta \mathcal{L}_k$, which is the starting point of the gradient-transmission analysis. (B) does *not* require $\mathcal{L}_k$ to depend only on the k-th pair in isolation: as in InfoNCE, each $\mathcal{L}_k$ may depend on the full batch, provided the loss still averages such per-anchor terms.

4 Weight Decay and Stationary Magnitude

Before the main theorem, let's discuss a simple illustrative example. In most models, the final encoder layer projects the backbone feature vector $h_i \in \mathbb{R}^d$ for sample i to the output embedding $z_i \in \mathbb{R}^k$. This layer is usually linear with weight matrix $W \in \mathbb{R}^{k\times d}$, so $z_i = W h_i$. The contrastive loss depends only on normalized vectors, so it is independent of the scaling of W: $\mathcal{L}(\alpha W) = \mathcal{L}(W)$ for any scalar $\alpha > 0$.

Differentiating $\mathcal{L}(\alpha W) = \mathcal{L}(W)$ with respect to α at $\alpha = 1$ gives[1]

$$\langle W, \nabla_W \mathcal{L}\rangle_F = \mathrm{Tr}(W^\top \nabla_W \mathcal{L}) = 0. \tag{2}$$

Under a gradient descent update $W' = W - \eta\nabla_W\mathcal{L}$, the squared Frobenius norm of the weights evolves as:

$$\|W'\|_F^2 = \|W - \eta\nabla_W\mathcal{L}\|_F^2 = \|W\|_F^2 - 2\eta\langle W, \nabla_W\mathcal{L}\rangle_F + \eta^2\|\nabla_W\mathcal{L}\|_F^2. \tag{3}$$
$$= \|W\|_F^2 + \eta^2\|\nabla_W\mathcal{L}\|_F^2, \tag{4}$$

And so the magnitude grows whenever we have a nonzero gradient step orthogonal to W. Qualitatively similar radial-growth effects due to tangential gradients were also noted in [27, 40, 6]. However, the main purpose of this paper is to derive a formula for the length of *individual* inputs.

Weight decay is usually applied to prevent norms from growing too large. Modern implementations such as AdamW do weight decay as a separate update from gradient step, and our discussion will also follow this convention. Let U_t denote the gradient step before decay (for SGD, $U_t = -\eta G_t$), and let Λ be the parameter-wise decay operator (diagonal, with entry λ_j for each parameter, with entries zero from parameters not decayed). Then,

$$\theta_{t+1} = (I - \eta\Lambda)(\theta_t + U_t) = \theta_t + U_t - \eta\Lambda\theta_t - \eta\Lambda U_t, \tag{5}$$

where the $-\eta\Lambda U_t$ cross-term is a slight deviation from the standard decoupled-weight-decay form $\theta_{t+1} = \theta_t + U_t - \eta\Lambda\theta_t$, but is $O(\eta^2)$ and dropped in the first-order expansion that follows.

After training, the task update does not necessarily vanish; instead, the process reaches a stationary distribution. Write $J_z = \partial f(x;\theta_t)/\partial\theta$ and expand Eq. (5) to first order in the parameter displacement:

$$\Delta z_t = z_{t+1} - z_t \approx \underbrace{J_z U_t}_{\Delta z_{\text{task}}} + \underbrace{(-\eta J_z \Lambda\theta_t)}_{\Delta z_{\mathrm{wd}}} + \underbrace{(-\eta J_z \Lambda U_t)}_{\text{cross term}}. \tag{6}$$

[1] $\|\cdot\|_F$ denotes the Frobenius norm.

Here Δz_{task} and Δz_{wd} are defined exactly as

$$\Delta z_{\text{task}}(x) := f(x;\, \theta_t + U_t) - f(x;\, \theta_t), \qquad \Delta z_{\text{wd}}(x) := f(x;\, \theta_{t+1}) - f(x;\, \theta_t + U_t),$$

so that $\Delta z_t = \Delta z_{\text{task}} + \Delta z_{\text{wd}}$ exactly. Given this we define the effective radial decay coefficient

$$\lambda_{\text{rad}}(x,t) := \frac{z_t(x)^\top J_z \Lambda \theta_t}{\|z_t(x)\|^2}. \tag{7}$$

Then

$$2z_t^\top \Delta z_{\text{wd}} \approx -2\eta\lambda_{\text{rad}}(x,t)\|z_t\|^2. \tag{8}$$

In practice, λ_{rad} would noisily fluctuate around a relatively stable mean value.

4.1 The Main Equation of Equilibrium Length

When training converges to a stationary distribution, the expected squared norm growth vanishes, and solving this equilibrium condition yields the following norm.

Theorem 1 (Stationary Embedding Norm)**.** *Let x be a fixed input whose embedding $z_t(x) = f(x;\theta_t)$ is trained under a loss function satisfying assumptions (A), (B) with learning rate $\eta > 0$ and decoupled weight decay. At each step t, given the parameters θ_t, the minibatch B_t is random; hence $\Delta z_{\text{task}}(x)$ is a random vector. When θ_t is sampled from the stationary distribution, define the per-step moments conditioned on the model state:*

$$\mu(x) := \mathbb{E}[\Delta z_{\text{task}}(x) \mid \theta_t], \qquad \tilde{\mathcal{V}}(x)^2 := \operatorname{Tr}(\operatorname{Cov}(\Delta z_{\text{task}}(x) \mid \theta_t)),$$

and let $\hat{z}(x) := z_t(x)/\|z_t(x)\|$ be its unit direction. Its radial drift $\tilde{\mathcal{D}}(x) := \hat{z}(x)^\top \mu(x)$, and total second-order heat $\Sigma(x)^2 := \|\mu(x)\|^2 + \tilde{\mathcal{V}}(x)^2$. The damping-normalised quantities

$$\mathcal{D}_\star(x) := \frac{\tilde{\mathcal{D}}(x)}{\eta\lambda_{\text{rad}}(x)}, \qquad \mathcal{V}_\star(x)^2 := \frac{\Sigma(x)^2}{\eta\lambda_{\text{rad}}(x)}, \tag{9}$$

Given $\|z_t(x)\| = R$, the expected squared-norm change conditioned on θ_t expands as

$$\mathbb{E}\big[\|z_{t+1}\|^2 - R^2 \mid \theta_t\big] = 2R\,\tilde{\mathcal{D}}(x) + \Sigma(x)^2 - 2\eta\lambda_{\text{rad}}(x)R^2. \tag{10}$$

When θ_t is drawn from the stationary distribution, $\mathbb{E}[\|z_{t+1}\|^2 \mid \theta_t] = \mathbb{E}[\|z_t\|^2 \mid \theta_t]$, so the left-hand side vanishes. Substituting the damping-normalised quantities and setting the right-hand side to zero yields the bi-quadratic balance for the stationary norm R_{eq}:

$$2R_{\text{eq}}^4 - 2\mathcal{D}_\star(x)\,R_{\text{eq}}^2 - \mathcal{V}_\star(x)^2 = 0, \tag{11}$$

whose unique positive solution is

$$R_{\text{eq}}^2 = \frac{\mathcal{D}_\star(x) + \sqrt{\mathcal{D}_\star(x)^2 + 2\mathcal{V}_\star(x)^2}}{2}. \tag{12}$$

What the variables measure. As we will see below, the theorem separates two scale-normalised task-update forces. $\tilde{\mathcal{V}}(x)^2$ is stochastic task heat: it is large when random minibatches send mutually inconsistent update votes. $\tilde{\mathcal{D}}(x)$ is structural radial drift.

The bi-quadratic Eq. (11) **yields three interpretable limits. Variance-dominated** concepts ($|\mathcal{D}_\star(x)| \ll \mathcal{V}_\star(x)$) satisfy $R_{\text{eq}} \approx (\mathcal{V}_\star(x)^2/2)^{1/4}$; if $\|\mu(x)\| \ll \tilde{\mathcal{V}}(x)$, then $R_{\text{eq}} \approx (\tilde{\mathcal{V}}(x)^2/(2\eta\lambda_{\text{rad}}(x)))^{1/4}$. Most concepts live here, especially those ambiguous: no stable neighborhood pulls them outward, but stochastic update heat still resists decay. **Outward-drift** concepts ($\mathcal{D}_\star(x) > 0$, $\mathcal{D}_\star(x)^2 \gg \mathcal{V}_\star(x)^2$) have $R_{\text{eq}}^2 \approx \mathcal{D}_\star(x)$ and $R_{\text{eq}} \approx (\tilde{\mathcal{D}}(x)/(\eta\lambda_{\text{rad}}(x)))^{1/2}$. Broad semantic hubs live here when many neighbors transmit updates that agree with the radial direction of x. **Inward-drift** concepts ($\mathcal{D}_\star(x) < 0$) suffer extra radial damping; in the strong inward limit $|\mathcal{D}_\star(x)|^2 \gg 2\mathcal{V}_\star(x)^2$, rationalizing the positive root gives $R_{\text{eq}}^2 \approx \mathcal{V}_\star(x)^2/(2|\mathcal{D}_\star(x)|) = (\|\mu(x)\|^2 + \tilde{\mathcal{V}}(x)^2)/(2|\tilde{\mathcal{D}}(x)|)$. Presumably, concepts with positive contexts spanning incompatible embedding directions live here.

4.2 NTK approximation and specificity

The NTK helps explain why $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{D}}$ measure what we claim above. Let $s_k := \|z_t(X_k)\|\nabla_{z_t(X_k)}\mathcal{L}_k$ be the scale-normalised source gradient for the k-th batch element[2], with $\bar{s}(X) = \mathbb{E}[s_k \mid X_k = X]$ and $\Sigma_s(X) = \mathrm{Cov}(s_k \mid X_k = X)$. Since $\nabla_z \mathcal{L} = s/\|z\|$ and neighbouring embeddings have similar norms $R \approx \|z_t(x)\| \approx \|z_t(X)\|$, the chain rule $\Delta z = J\Delta\theta$ gives $\Delta z_{\text{task}} \approx -\frac{\eta}{R}\sum_k \mathcal{K}(x, X_k)s_k$. For i.i.d. batch slots, taking moments (see Appendix E for derivation):

$$\mu(x) \approx -\frac{\eta B}{R}\,\mathbb{E}_X[\mathcal{K}(x,X)\bar{s}(X)]\,, \tag{13}$$

$$\tilde{\mathcal{V}}(x)^2 \approx \frac{\eta^2 B}{R^2}\,\mathrm{Tr}\big(\mathbb{E}_X\left[\mathcal{K}(x,X)\Sigma_s(X)\mathcal{K}(x,X)^\top\right] + \mathrm{Cov}_X(\mathcal{K}(x,X)\bar{s}(X))\big)\,. \tag{14}$$

When the conditional mean is small, the variance proxy reduces to

$$\tilde{\mathcal{V}}(x)^2 \approx \frac{\eta^2 B}{R^2}\,\mathbb{E}_X\left[s(X)^\top \mathcal{K}(x,X)^\top \mathcal{K}(x,X)s(X)\right]\,. \tag{15}$$

Thus $\tilde{\mathcal{V}}$ is large when the "NTK neighborhood" of x (that is, input treated as similar to x by the model with parameter θ_t) contains mutually inconsistent information according to training data labels. Generic concepts, polysemous inputs, and perceptually ambiguous images receive diverse tangent kicks; specific concepts receive small, repeatable kicks because their positives concentrate around one direction. In the regime $\tilde{\mathcal{V}} \propto R^2$, norm becomes a proxy for specificity.

Radial drift depends on the sign of those transmitted votes. Since the source gradient is tangent to the unit sphere at X, only the tangent projection

$$a_K(x,X) = P^{\perp}_{\hat{z}(X)}\mathcal{K}(x,X)^\top \hat{z}(x)$$

can change the length of $z(x)$. If $\varphi_K(x,X)$ is the angle between $\mathcal{K}(x,X)^\top\hat{z}(x)$ and $\hat{z}(X)$, and $\psi_K(x,X)$ is the angle between $a_K(x,X)$ and $-\bar{s}(X)$, then

$$\tilde{\mathcal{D}}(x) \approx \frac{\eta B}{R}\,\mathbb{E}_X\left[\|\mathcal{K}(x,X)^\top\hat{z}(x)\|\sin\varphi_K(x,X)\|\bar{s}(X)\|\cos\psi_K(x,X)\right]\,. \tag{16}$$

Positive drift requires both strong transmission and positive alignment. A frequent but NTK-isolated input has small $\|a_K(x,X)\|$ and stays variance-dominated. A genuine hub has large $\|a_K(x,X)\|$ for many neighbors and $\cos\psi_K > 0$ on average, so neighborhood updates lengthen its vector. A boundary concept can have equally strong transmission but $\cos\psi_K < 0$, so the same NTK coupling compresses the norm. Figure 1 illustrates the geometry. The source gradient $\bar{s}(X)$ is tangent to the sphere because the loss depends only on the normalised embedding $\hat{z} = z/\|z\|$; scale invariance gives $\nabla_z\mathcal{L}\cdot z = 0$.

4.3 Implications for Norm-Aware Similarity Scoring

The task-heating term $\mathcal{V}_\star(x)$ encodes neighborhood uncertainty: low $\mathcal{V}_\star$ marks a specific, well-defined concept, so a match between two low-$\mathcal{V}_\star$ inputs carries more information at the same cosine angle than a match between two high-$\mathcal{V}_\star$ inputs.

At inference, it is possible to compute $\mathcal{V}_\star$ at some cost but R is already present. Theorem 1 makes R a possible proxy for $\mathcal{V}_\star$ when variance dominates. This then motivates a norm-discounted similarity score

$$\mathrm{score}(q,d) = \cos\theta(q,d)\cdot\|q\|^{-\gamma}\cdot\|d\|^{-\gamma}, \tag{17}$$

where $\gamma > 0$ discounts high-norm, high-uncertainty embeddings. Pure cosine corresponds to $\gamma = 0$ and discards radial information; the dot product corresponds to $\gamma = -1$, amplifying the embeddings this argument says to downweight.

[2]The notation $\nabla_{z_t(X_k)}\mathcal{L}_k$ is a slight abuse: $\mathcal{L}_k$ depends on the full batch through the softmax denominator, so this is not a partial derivative in the strict sense but the gradient of the total loss with respect to the embedding $z_t(X_k)$, holding other embeddings fixed.

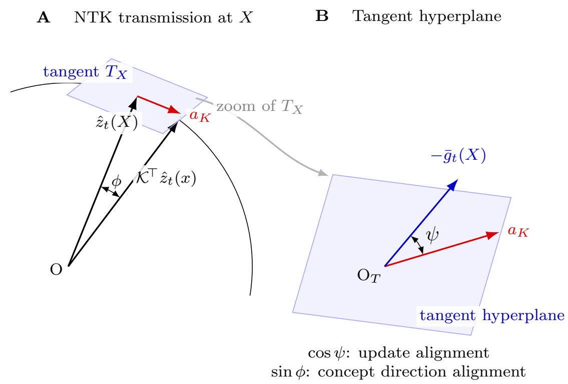

Figure 1: Sphere geometry of the radial drift term. At a training sample X, the transported radial direction $\mathcal{K}(x, X)^\top \hat{z}(x)$ has tangent projection a_K in $T_X\mathbb{S}^{d-1}$. The source update $-\bar{s}(X)$ also lies in this tangent space. Its angle ψ_K with a_K sets the sign of the radial vote: $\cos\psi_K > 0$ pushes x outward, while $\cos\psi_K < 0$ pushes it inward.

4.4 The Optimizer Artifacts of Frequency Scaling (AdamW vs. SGD)

For simplicity and due to length limit of the paper, we derive the main equation for SGD, but the core equilibrium principle is independent of optimizer choice, and as we will see in experiments, even if we compute $\tilde{\mathcal{V}}$ using SGD on a model trained with some other optimizer, the results are still informative. However, the math does change: for example, momentum optimizers rescales stochastic heating by $(1-\beta)/(1+\beta)$ without changing which concepts receive larger $\tilde{\mathcal{V}}$. Derivation of this can be found in Appendix G. AdamW combines this momentum rescaling with the decoupled weight decay already analyzed in Section 4, so its net effect on $\tilde{\mathcal{V}}$ follows from multiplying the two factors.

5 Experiments

We will verify the main equation's predictions, before showing how this can potentially be used to improve similarity scoring and explain some effects.

5.1 Verifying the main equation: equilibrium scaling with λ

λ Scaling. The variance-dominated limit predicts $R^2_{\text{eq}} \propto (\tilde{\mathcal{V}}^2/\lambda)^{1/2}$, coupling equilibrium norm to task heat and weight decay. To verify this on several architectures, we firstly fine-tune `distilbert-base-uncased` [23] on a synthetic dataset containing artificial concepts, each paired with 100 positive contexts consisting of diverse sentence variations. Using this set, we train the model (initialized with the original weights of DistilBert) with SGDW over a range of decay values λ, training each run to norm convergence. Not surprisingly, R decreases as λ increases, as shown in the left panel of Figure 2. The more interesting right panel of the figure shows that the predicted scaling also holds: after log-log linear fitting of R^2 against $\tilde{\mathcal{V}}^2/\lambda$, we see a gradient very close to 1, verifying the coupled relationship introduced by the main equation.

Verifying the main equation. To verify Theorem 1, we fine-tune `distilbert-base-uncased` [23] on MultiNLI [34] using SGDW ($\eta = 0.01$, $w = 0.01$). We measure pooled sentence-embedding norms at convergence and estimate $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{D}}$ via a Monte Carlo estimator. Because λ_{rad} varies across vocabulary positions, we take suitable averages. Details are in Appendix H.

Figure 3 presents the results. Panel (a) tests the full equation: plotting the predicted norm against the observed norm yields a log-log gradient of 0.88 with Pearson correlation $r = 0.97$. The gradient $0.88 < 1$ and the y-intercept reflect a systematic under-prediction that stems from residual non-stationarity: although the model has reached convergence by any practical standard (training loss is flat, gradients are small), the embedding norms are still slowly decreasing over the measurement

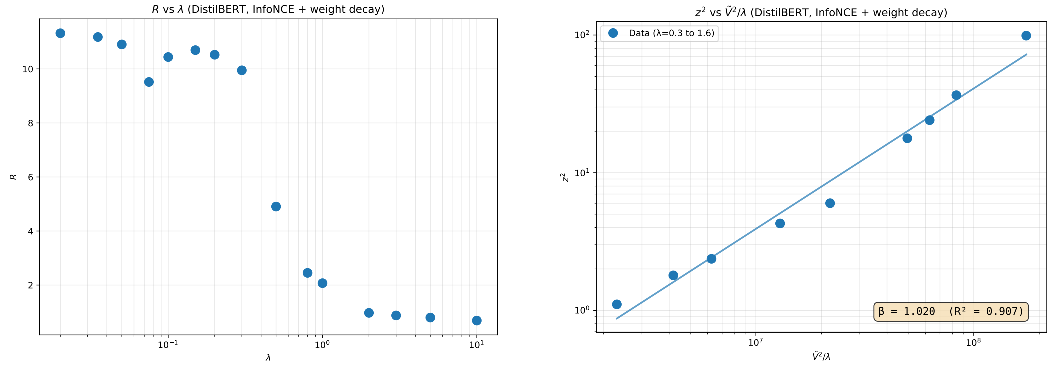

Figure 2: Left: average R across tokens against λ on log scale. Right: log-log fitting of R^2 against $\tilde{\mathcal{V}}^2/\lambda$ confirming the theoretical scaling.

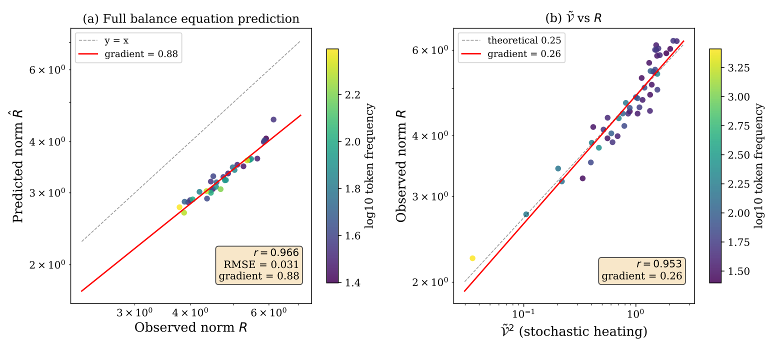

Figure 3: **(a)** Log-log scatter of predicted vs. observed embedding norm from the full balance equation (Eq. (12)). Each point is a vocabulary position; color encodes token frequency. The fitted gradient is 0.88 ($r = 0.97$), confirming that $\tilde{\mathcal{V}}$ and λ_{rad} capture the dominant mechanism governing equilibrium norms. The systematic deviation from $y = x$ reflects residual non-stationarity in the measurement window. **(b)** $\tilde{\mathcal{V}}$ vs. $\|z_{\text{eq}}\|$ on log-log axes. The fitted gradient 0.26 matches the theoretical $+0.25$ prediction (dashed), confirming that stochastic heating $\tilde{\mathcal{V}}$ is the primary driver of embedding length.

window. This biases the prediction slightly downward, yet it does not weaken the verification — the high correlation confirms the balance equation captures the dominant mechanism, and the bias magnitude is quantitatively consistent with the measured residual drift.

Because $\tilde{\mathcal{D}}$ and λ_{rad} are noisy to estimate precisely (for $\tilde{\mathcal{D}}$, estimators disagree on sign for 85% of tokens; for λ_{rad}, the median within-token CV across 20 repeated measurements is 0.64), it is worth examining whether the noise-dominated limit, which only involves $\tilde{\mathcal{V}}$, holds in practice. When $\tilde{\mathcal{D}} \approx 0$, Eq. (12) reduces to $\|z_{\text{eq}}\| \approx (\tilde{\mathcal{V}}^2/2\eta\lambda_{\text{rad}})^{1/4}$, predicting a log-log slope of 0.25 between $\tilde{\mathcal{V}}$ and $\|z_{\text{eq}}\|$. Panel (b) confirms this: the measured slope is 0.26 ($r = 0.95$). In contrast to $\tilde{\mathcal{D}}$ and λ_{rad}, the between-estimator correlation for Σ^2 (from which $\tilde{\mathcal{V}}^2$ is derived) is $r = 0.91$, and $\tilde{\mathcal{V}}^2$ predicts the norm at $r = 0.95$ across tokens. This robustness makes $\tilde{\mathcal{V}}$ a practical tool for analyzing embedding geometry across diverse models without retraining, as we will see in later subsections.

In Appendix A, we repeat this analysis for CLIP [21] text embeddings and MiniLM-L12 [32] *without* fine-tuning, but we see similar effect (but to a weaker extent). This shows that although the quantities technically depend on optimizers and training data, the effect we observe is still very robust.

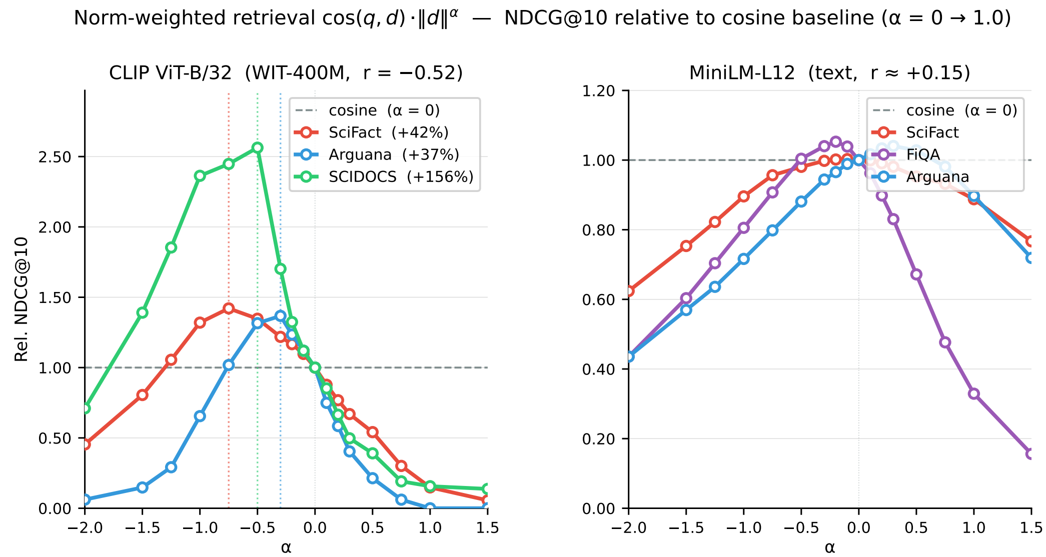

Figure 4: Norm-weighted BEIR retrieval: $\text{score}(q, d) = \cos(q, d) \cdot \|d\|^{\alpha}$ ($\alpha = -\gamma$). NDCG@10 normalised to cosine baseline ($\alpha = 0 \to 1.0$). **Left:** CLIP ViT-B/32 gains 37–156% at negative α across three datasets. **Right:** MiniLM-L12 shows no benefit at any α.

5.2 Norm as a proxy for semantic specificity and model uncertainty

The theory identifies $\tilde{\mathcal{V}}$ with the specificity of its corresponding information or concept. Section 4 suggests that the norm is also a proxy for inverse specificity. In Appendix C, we empirically investigate the relationship of both $\tilde{\mathcal{V}}$ and R to semantic specificity, showing how stochastic heating $\tilde{\mathcal{V}}$ concentrates on examples the model finds ambiguous, and how the norm tracks specificity at the answer level.

5.3 Norm-aware scoring might improve retrieval

Section 4.3 proposed the norm-discounted similarity score

$$\text{score}(q, d) = \cos\theta(q, d) \cdot \|q\|^{-\gamma} \cdot \|d\|^{-\gamma}.$$

For a fixed query, $\|q\|^{-\gamma}$ is constant and cannot affect the within-query ranking, so retrieval experiments use the equivalent document-weighted score $\cos(q, d) \cdot \|d\|^{\alpha}$ with $\alpha = -\gamma$. The symmetric score remains relevant for pairwise calibration or classification across varying queries. In retrieval settings that prefer specific answers, $\alpha < 0$ discounts high-norm, generic documents, whereas the dot product corresponds to $\alpha = 1$ and boosts them.

Figure 4 shows a sweep on BEIR [25], a heterogeneous benchmark of information retrieval datasets, with the document exponent α.

We repeat the comparison on a LAION text–text retrieval benchmark with DistilBERT [23] and E5 [28] embeddings; the qualitative improvement from norm discounting persists across models, confirming that the benefit does not depend on the multimodal setting. Appendix B reports a depth-stratified TREC CAR stress test, where CLIP variants benefit most from negative α at deeper Wikipedia heading levels while text-only models stay near cosine.

6 Conclusion

This paper provides a theoretical mechanism explaining why embedding norms in contrastive models correlate with semantic properties despite being discarded by the training loss. Our central result, Theorem 1, presents a bi-quadratic equilibrium equation that balances radial weight decay against two update statistics: structural radial drift ($\tilde{\mathcal{D}}$) and stochastic heating ($\tilde{\mathcal{V}}$). In heat-controlled settings,

$\tilde{\mathcal{V}} \propto R^2$, making the norm a free, zero-shot proxy for semantic specificity and labeling uncertainty. Building on this insight, we demonstrate that a norm-aware scoring function that discounts large norms consistently improves retrieval performance on BEIR and TREC CAR, requiring no additional training or supervision.

7 Limitations

The main caveat is that the loss-function assumptions are not entirely consistent with mean pooling: scaling one token state before pooling can change the pooled direction, so the current equation is not quantitatively exact for mean-pooled embeddings. Mean pooling certainly has similar qualitative radial-heating and norm-specificity behavior, suggesting that a pooling-aware generalization would substantially broaden the theory. The same applies to optimizer choice: as discussed above, the derivation uses SGD-style updates, while momentum and AdamW change the effective drift and heat. These generalizations are natural future work. Finally, $\tilde{\mathcal{D}}$ and λ_{rad} are noisy to estimate, whereas $\tilde{\mathcal{V}}$ is stable and predicts R well in our experiments.

8 Impact Statement

This work develops a theoretical framework for understanding embedding norm dynamics in contrastive learning and derives a practical norm-aware similarity score. The primary applications are retrieval and zero-shot classification using existing pretrained models; no new models or datasets are released. We anticipate no direct negative societal impacts. Improved retrieval quality could modestly benefit downstream systems that rely on semantic search, with the usual caveats that such systems inherit biases present in pretraining data. The theory itself is foundational: it may inform future training procedures, regularization choices, and evaluation protocols for embedding models, potentially improving their reliability and interpretability.

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A Additional Model Verifications

We repeat the $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{D}}$ measurement protocol from Section 5.1 **without fine-tuning** on two additional architectures—OpenAI CLIP ViT-B/32 (vision-language, 151M) and sentence-transformers/all-MiniLM-L12-v2 (text-only, 33M). For each model we select $n = 96$ probe tokens spanning the full range of observed embedding norms, draw $B = 48$ random minibatches, and compute the Monte Carlo estimators described in Appendix H.

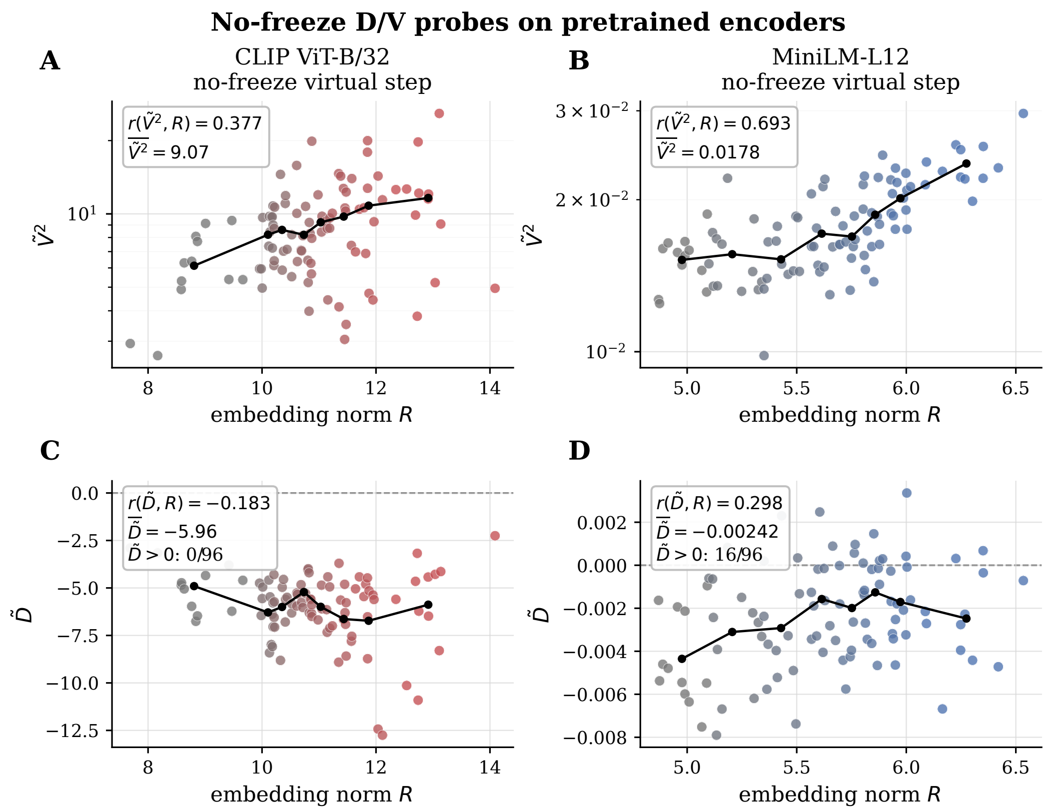

Figure 5: $\tilde{\mathcal{V}}^2$ **and** $\tilde{\mathcal{D}}$ **measurements.** Top row: per-token $\tilde{\mathcal{V}}^2$ and $\tilde{\mathcal{D}}$ values sorted by observed embedding norm for CLIP ViT-B/32 (left) and MiniLM-L12 (right). Bottom row: log-log relationship between $\tilde{\mathcal{V}}^2$ and norm, and $\tilde{\mathcal{D}}$ vs. $\tilde{\mathcal{V}}^2$ scatter.

Table 1: $\tilde{\mathcal{D}}$ and $\tilde{\mathcal{V}}^2$ summary statistics ($n = 96$ probes each). Means reported $\pm$ SD. $\tilde{\mathcal{D}} < 0$ indicates systematic inward radial drift. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$ (two-sided t-test for Pearson r, d.f. $= 94$).

Model	$\langle\tilde{\mathcal{V}}^2\rangle$	$\langle\tilde{\mathcal{D}}\rangle$	$r(\tilde{\mathcal{V}}^2, R)$	$r(\tilde{\mathcal{D}}, R)$	$r(\tilde{\mathcal{D}}, \tilde{\mathcal{V}}^2)$
CLIP ViT-B/32	9.07 ± 4.09	-5.96 ± 1.79	$+0.377^{***}$	-0.183	-0.580^{***}
MiniLM-L12	0.0178 ± 0.0037	-0.0024 ± 0.0024	$+0.693^{***}$	$+0.298^{**}$	$+0.158$

Two findings stand out. First, MiniLM-L12 has negligible drift: $\langle\tilde{\mathcal{D}}\rangle \approx 0$, the fraction of positive-drift tokens is near chance (16/96), and $\tilde{\mathcal{V}}^2$ predicts the norm at $r = 0.693$. This matches the DistilBERT result from Section 5.1.

Second, CLIP ViT-B/32 exhibits *systematic inward drift*: all 96 probes have $\tilde{\mathcal{D}} < 0$ (mean -5.96), and the drift is anti-correlated with stochastic heat ($r = -0.580$). This places CLIP in the inward-drift regime of Section 4.1, where structural drift acts as additional radial damping. The log-log slope of $\tilde{\mathcal{V}}^2$ vs. R is 1.47, intermediate between the pure variance-dominated prediction (0.25) and the drift-dominated prediction, consistent with both terms contributing to the equilibrium.

Table 2: NDCG@10 gain (%) from norm-weighted retrieval on TREC CAR, by Wikipedia heading depth. Positive gain means a negative α (norm discounting) improves over cosine. Overall column shows gain and the optimal α across all 1,860 queries.

Model	Type	Overall (best α)	d=1 (22q)	d=2 (1012q)	d=3 (685q)	d=4 (134q)
		Vision-language with projection head				
CLIP ViT-B/32	VL+proj	**+32.3%** (-0.75)	+19.6%	+35.2%	+27.1%	+37.9%
CLIP ViT-B/16	VL+proj	**+37.7%** (-0.75)	+35.6%	+39.1%	+33.4%	+43.3%
CLIP ViT-L/14	VL+proj	**+44.3%** (-1.25)	+18.7%	+41.1%	+48.2%	+62.2%
OpenCLIP ViT-B/32	VL+proj	+9.6% (-1.00)	+23.5%	+7.1%	+14.3%	+16.6%
		VL without projection head (ablation)				
CLIP mean-pool	VL$-$proj	+7.9%* (+0.50)	+5.2%	+8.4%	+6.6%	+16.8%
		Text-only models				
E5-base-v2	Text	+0.5% (-0.10)	+3.3%	0.0%	+1.1%	+1.6%
BGE-large-v1.5	Text	0.0% (+0.00)	+1.9%	0.0%	+0.2%	+1.1%
SBERT-MPNet	Text	+1.3% (-0.30)	+12.4%	+1.6%	+1.3%	+1.5%
MiniLM-L6	Text	+1.4% (-0.10)	+4.4%	+1.4%	+2.3%	+0.3%
MiniLM-L12	Text	+0.7% (-0.20)	+3.2%	+1.0%	+0.9%	+1.5%
DistilBERT	Text	+1.5% (-0.10)	+3.3%	+1.5%	+0.8%	+7.9%
ALBERT-small	Text	0.0% (+0.00)	+5.1%	+0.3%	0.0%	+2.0%

*Gain uses *positive* α, i.e. boosting high-norm (generic) documents — the opposite mechanism from VL+proj models, confirming the projection head is necessary for the specificity-norm signal.

The contrast between CLIP and MiniLM explains *why* the norm-specificity relationship (Section 5.2) and norm-aware retrieval gains (Section 5.3) appear in vision-language models but not in text-only models. The inward drift in CLIP creates a regime where $\tilde{\mathcal{D}}$ and $\tilde{\mathcal{V}}^2$ are anti-correlated, producing a negative norm-drift correlation that amplifies the norm-specificity signal. In text-only models, where $\tilde{\mathcal{D}} \approx 0$, norm is driven purely by $\tilde{\mathcal{V}}^2$ and the specificity signal, while present, is weaker and does not translate into retrieval gains.

B Depth-Stratified TREC CAR Retrieval

To test whether norm discounting preferentially benefits queries at deeper specificity levels, we evaluate on TREC CAR [5] (Complex Answer Retrieval), where each of $1{,}860$ test queries is annotated with Wikipedia heading depth—page (depth 1, 22 queries), section (depth 2, $1{,}012$ queries), subsection (depth 3, 685 queries), and subsubsection (depth 4, 134 queries). Deeper headings correspond to more specific subtopics. We sweep the document-norm exponent α in $\mathrm{score}(q, d) = \cos(q, d) \cdot \|d\|^{\alpha}$ and measure NDCG@10 gain over cosine ($\alpha = 0$) at each depth. We compare CLIP [21] variants and OpenCLIP [8] ViT-B/32 against text-only models (E5 [28], BGE-large-v1.5 [35], SBERT-MPNet [22], MiniLM-L6 and -L12 [32], DistilBERT [23], ALBERT-small [13]).

Table 2 and Figure 6 show a benchmark-specific split. All four vision-language models with projection heads gain substantially (+9.6% to +44.3%), and three of four select increasingly negative α as query depth increases from section (d=2) to subsubsection (d=4). Removing the projection head (CLIP mean-pool) flips the best exponent positive and removes the depth gradient. All seven text models stay near cosine ($\leq 1.5\%$ gain), with no systematic α pattern. These retrieval results stress-test, rather than repeat, the model-regime distinction measured in Section 5.1.

C Norm as a proxy for semantic specificity and model uncertainty

The theory identifies $\tilde{\mathcal{V}}$ with the specificity of its corresponding information or concept. Section 4 suggests that the norm is also a proxy for inverse specificity. In the following experiments, we empirically investigate the relationship of both $\tilde{\mathcal{V}}$ and R to semantic specificity.

CIFAR-10H $\tilde{\mathcal{V}}$ multi-model experiment. We test whether $\tilde{\mathcal{V}}$ measures specificity and uncertainty on CIFAR-10H [19], which provides human soft-label distributions for 10000 CIFAR-10 images. We

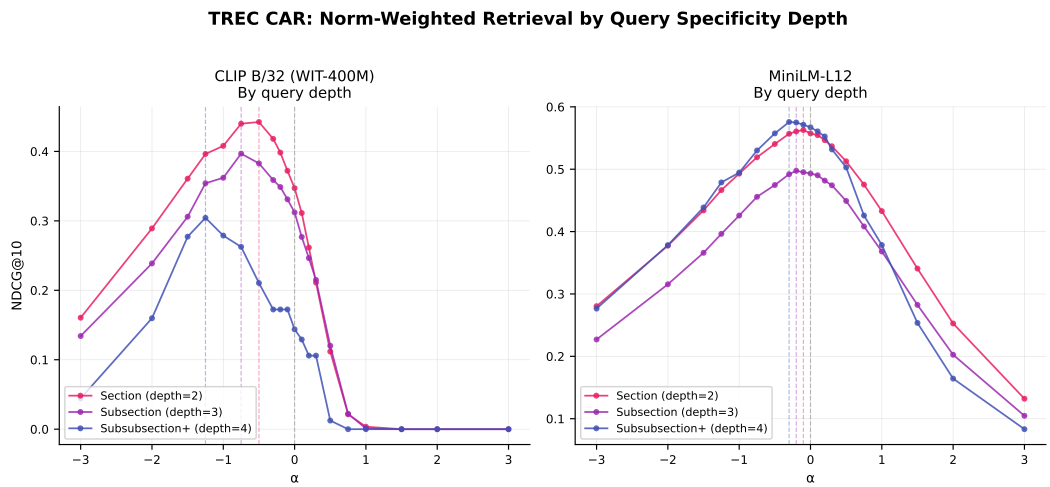

Figure 6: TREC-CAR depth-stratified norm-weighted retrieval. Each colour represents a query specificity level (depth 2 = section, depth 3 = subsection, depth 4+ = subsubsection). CLIP benefits most from negative α at deeper heading levels; MiniLM-L12 shows no depth gradient.

Table 3: $\tilde{\mathcal{V}}^2$ **correlations across 14 vision-language models on CIFAR-10H.** $\tilde{\mathcal{V}}^2$ tracks model-internal uncertainty ($r(\tilde{\mathcal{V}}^2, H_{\text{model}})$) more strongly than human annotator uncertainty ($r(\tilde{\mathcal{V}}^2, H_{\text{human}})$). SigLIP models (marked †) use sigmoid rather than contrastive loss and show consistently weaker or negative correlations.

Model	$r(\tilde{\mathcal{V}}^2, H_{\text{model}})$	$r(\tilde{\mathcal{V}}^2, H_{\text{human}})$	$r(\tilde{\mathcal{V}}^2, \|z\|)$
CLIP ViT-B/32	$+0.234^{***}$	$+0.050$	$+0.350^{***}$
CLIP ViT-B/16	$+0.205^{***}$	$+0.147^{*}$	$+0.179^{**}$
CLIP ViT-L/14@336px	$+0.075$	$+0.098$	$+0.090$
CLIP ViT-L/14	$+0.052$	$+0.010$	$+0.364^{***}$
OpenCLIP ViT-B/32	$+0.319^{***}$	$+0.210^{***}$	$+0.436^{***}$
OpenCLIP ViT-H/14	$+0.177^{**}$	$+0.178^{**}$	$+0.126^{*}$
OpenCLIP ViT-L/14	$+0.123^{*}$	$+0.236^{***}$	$+0.259^{***}$
MetaCLIP ViT-B/16	$+0.307^{***}$	$+0.087$	$+0.495^{***}$
MetaCLIP ViT-H/14	$+0.204^{**}$	$+0.070$	$+0.356^{***}$
MetaCLIP ViT-B/32	$+0.190^{**}$	$+0.161^{**}$	$+0.397^{***}$
MetaCLIP ViT-L/14	$+0.114$	$+0.192^{**}$	$+0.208^{***}$
SigLIP Base Patch16/384†	$+0.030$	$+0.043$	$+0.078$
SigLIP Large Patch16/384†	-0.029	$+0.099$	-0.172^{**}
SigLIP SO400M Patch14/384†	-0.066	-0.026	$+0.192^{**}$

† SigLIP uses sigmoid loss instead of contrastive. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$ (two-sided t-test, $n = 256$, d.f. $= 254$).

measure $\tilde{\mathcal{V}}$ for 256 norm-stratified probe images via 48 virtual contrastive SGD steps, across 14 vision-language models spanning OpenAI CLIP [21], OpenCLIP [8], MetaCLIP [36], and SigLIP [37]. For each probe we compute H_{model} (zero-shot softmax entropy) and H_{human} (entropy of the human soft-label distribution). Table 3 reports Pearson correlations. Across all 11 standard contrastive models, $r(\tilde{\mathcal{V}}^2, H_{\text{model}})$ is uniformly positive, confirming that stochastic heating tracks model-internal uncertainty. $r(\tilde{\mathcal{V}}^2, H_{\text{human}})$ is also positive but consistently smaller, and $r(\tilde{\mathcal{V}}^2, \|z\|)$ is largest of all—the hierarchy $\|z\| > H_{\text{model}} > H_{\text{human}}$ holds across models. Three SigLIP variants (sigmoid rather than contrastive loss) produce near-zero or negative correlations throughout, confirming the signal is specific to contrastive training. The sign consistency across 11 independently trained standard models argues against a spurious explanation.

Norm-specificity calibration via general/specific Q/A pairs. To further probe whether norm tracks specificity at the answer level (rather than the concept level), we construct a synthetic Q/A dataset

Table 4: Embedding norm and cosine similarity: specific vs. general answers under the repeated-general protocol ($n = 1060$ pairs). $z = \sqrt{n}\,\bar{\Delta}/\sigma_{\Delta}$ (paired z-statistic); %Norm↓ = pairs where $\|\text{specific}\| < \|\text{general}_{\times 2}\|$; %Cos↓ = pairs where $\cos(q, \text{specific}) < \cos(q, \text{general})$. Negative z means specific answers have reliably smaller norms as predicted. Cosine differences are uniformly near zero, confirming equal directional relevance.

Model	z	%Norm↓	%Cos↓
Encoder (mean-pool)			
MiniLM-L3	−45.0	93.5%	66.9%
E5-Large-v2	−52.7	96.1%	85.4%
SBERT (all-mpnet)	−20.1	73.2%	64.6%
BGE-M3	−12.5	65.1%	74.0%
Vision-language (projection head)			
CLIP ViT-B/32	−6.3	60.6%	87.1%
Decoder (last-token)			
Qwen3-0.6B	−14.4	67.6%	59.6%
GTE-Qwen2-1.5B	+0.0	51.3%	48.7%
Jina-v3	+0.2	51.6%	70.8%
SFR-Mistral	+36.4	11.9%	71.3%
E5-Mistral-7B	+31.0	18.0%	63.9%

using a two-stage LLM pipeline. For each item, a generator LLM (DeepSeek [4]) produces a question, a general answer (broad background, definitions), and a specific answer (concrete mechanistic or quantitative details, same relevance). Another judge LLM validates each triple, requiring both answers to be equally relevant and correct, the specific to entail the general but not vice versa, and the specific to add concrete information. The final dataset contains 1,060 validated triples spanning 17 domains.

We extract unnormalized embeddings for each answer across 8 models: mean-pool encoders MiniLM-L3 [32], E5-Large-v2 [28], SBERT (all-mpnet) [22], BGE-M3 [2]; vision-language CLIP ViT-B/32 [21]; and decoder models Qwen3-0.6B [20], GTE-Qwen2-1.5B [39], Jina-v3 [24], SFR-Mistral [17], E5-Mistral-7B [29]. To control for the differing text lengths of the two answers, we apply the *repeated-general protocol*: the general answer text is concatenated with itself before encoding, matching the approximate token count of the specific answer. For mean-pooled models this leaves the embedding nearly unchanged; for decoder models the 2× repetition alters the terminal context, so the comparison is less direct. The theory predicts $\|\text{specific}\| < \|\text{general}\|$ because the specific answer's narrower semantic content induces lower $\tilde{\mathcal{V}}$.

Table 4 reports the results. All encoder models show specific < general, with z-statistics exceeding 6 standard errors in every case. As a control, we also compute $\cos(q, \text{specific}) - \cos(q, \text{general})$: the difference is uniformly near zero (all $|\Delta| < 0.05$), confirming that both answers are equally relevant in direction space and that the norm effect is not a relevance confound. Decoder models (4 of 5) either invert or null the pattern, with Qwen3-0.6B as the sole decoder showing the encoder direction, consistent with their $\tilde{\mathcal{D}} \approx 0$ measurement from Section 5.1.

D Derivation of the Stationary Norm

Main equation. Fix an input x, abbreviate $z_t = z_t(x)$, $R = \|z_t\|$, and $\hat{z}_t = z_t/R$. Let $\Delta z_{\text{task}}(x)$ denote the task update before weight decay, and write its conditional mean and fluctuation as

$$\bar{\Delta}_t(x) := \mathbb{E}[\Delta z_{\text{task}}(x) \mid \theta_t], \qquad \varepsilon_t(x) := \Delta z_{\text{task}}(x) - \bar{\Delta}_t(x), \qquad \mathbb{E}[\varepsilon_t(x) \mid \theta_t] = 0. \tag{18}$$

The squared-norm expansion is

$$\mathbb{E}[\|z_t + \Delta z_{\text{task}}\|^2 - \|z_t\|^2 \mid \theta_t] = 2 z_t^\top \bar{\Delta}_t + \|\bar{\Delta}_t\|^2 + \operatorname{Tr}(\operatorname{Cov}(\Delta z_{\text{task}} \mid \theta_t)). \tag{19}$$

Define $\lambda_{\text{rad}}(x, t)$ as the effective radial decay rate of $\|z_t\|^2$ under weight decay:

$$\|z_t + \Delta z_{\text{wd}}(x)\|^2 - \|z_t\|^2 =: -2\eta \lambda_{\text{rad}}(x, t) R^2. \tag{20}$$

Define the rescaled task update $\Delta\tilde{z}_{\text{task}}(x) := R \cdot \Delta z_{\text{task}}(x)$ and its moments

$$\mu(x) := \mathbb{E}[\Delta\tilde{z}_{\text{task}}(x) \mid \theta_t], \tag{21}$$
$$\tilde{\mathcal{D}}(x) := \hat{z}_t(x)^\top \mu(x), \tag{22}$$
$$\tilde{\mathcal{V}}(x)^2 := \mathrm{Tr}(\mathrm{Cov}(\Delta\tilde{z}_{\text{task}}(x) \mid \theta_t)), \tag{23}$$

and $\Sigma(x)^2 := \|\mu(x)\|^2 + \tilde{\mathcal{V}}(x)^2$. By definition, $\bar{\Delta}_t = \mu(x)/R$, so $z_t^\top \bar{\Delta}_t = \tilde{\mathcal{D}}(x)$ and $\|\bar{\Delta}_t\|^2 + \mathrm{Tr}(\mathrm{Cov}(\Delta z_{\text{task}} \mid \theta_t)) = \Sigma(x)^2/R^2$ exactly. The expected squared-norm change, combining the task expansion with the weight-decay contraction, expands exactly to

$$\mathbb{E}\big[\|z_{t+1}\|^2 - \|z_t\|^2 \mid \theta_t\big] = -2\eta\lambda_{\text{rad}}(x,t)R^2 + 2\tilde{\mathcal{D}}(x) + \frac{\Sigma(x)^2}{R^2} + \underbrace{2\bar{\Delta}_t^\top \Delta z_{\text{wd}}(x)}_{O(\eta^2),\ \text{dropped}}. \tag{24}$$

Dropping the $O(\eta^2)$ cross term,

$$\mathbb{E}\big[\|z_{t+1}\|^2 - \|z_t\|^2 \mid \theta_t\big] \approx -2\eta\lambda_{\text{rad}}(x)R^2 + 2\tilde{\mathcal{D}}(x) + \frac{\Sigma(x)^2}{R^2}. \tag{25}$$

At stationarity of $\|z_t\|^2$, the left-hand side vanishes. With

$$\mathcal{D}_\star(x) := \frac{\tilde{\mathcal{D}}(x)}{\eta\lambda_{\text{rad}}(x)}, \qquad \mathcal{V}_\star(x)^2 := \frac{\Sigma(x)^2}{\eta\lambda_{\text{rad}}(x)}, \tag{26}$$

setting the right-hand side to zero and dividing by $\eta\lambda_{\text{rad}}(x)$ yields

$$-2R^2 + 2\mathcal{D}_\star(x) + \frac{\mathcal{V}_\star(x)^2}{R^2} = 0. \tag{27}$$

Multiplying by R^2 gives the bi-quadratic

$$2R^4 - 2\mathcal{D}_\star(x)R^2 - \mathcal{V}_\star(x)^2 = 0, \tag{28}$$

whose positive root is Theorem 1.

NTK approximation. At step t, the minibatch update satisfies $\Delta\theta \approx -\eta \sum_{k\in B_t} \nabla_\theta \mathcal{L}_k$. The chain rule sends this update to the embedding of the fixed input x:

$$\Delta z_{\text{task}}(x) \approx -\eta \sum_{k\in B_t} J_z(x) J_z(X_k)^\top \nabla_{z_t(X_k)} \mathcal{L}_k. \tag{29}$$

The matrix $\mathcal{K}(x, X_k) = J_z(x)J_z(X_k)^\top$ is the Neural Tangent Kernel [9]. Since the loss depends on normalized embeddings, its source gradient is tangent and inverse-length scaled. Define

$$s_k := \|z_t(X_k)\|\nabla_{z_t(X_k)}\mathcal{L}_k, \qquad \nabla_{z_t(X_k)}\mathcal{L}_k = \frac{s_k}{\|z_t(X_k)\|}, \qquad \hat{z}_t(X_k)^\top s_k = 0. \tag{30}$$

If the relevant NTK neighborhood has norms comparable to $R = \|z_t(x)\|$, then

$$\Delta z_{\text{task}}(x) \approx \frac{1}{R}\Delta\tilde{z}_{\text{task}}(x), \qquad \Delta\tilde{z}_{\text{task}}(x) := -\eta \sum_{k\in B_t} \mathcal{K}(x, X_k) s_k. \tag{31}$$

Thus the abstract unit-norm kick in the main equation is the NTK-transmitted sum of tangent source gradients.

Interpretable NTK moments. Let $\bar{s}(X) := \mathbb{E}[s_k \mid X_k = X]$ and $\Sigma_s(X) := \mathrm{Cov}(s_k \mid X_k = X)$. For i.i.d. batch slots,

$$\mu(x) \approx -\eta B\, \mathbb{E}_X[\mathcal{K}(x,X)\bar{s}(X)], \tag{32}$$
$$\tilde{\mathcal{V}}(x)^2 \approx \eta^2 B\, \mathrm{Tr}\big(\mathbb{E}_X\left[\mathcal{K}(x,X)\Sigma_s(X)\mathcal{K}(x,X)^\top\right] + \mathrm{Cov}_X(\mathcal{K}(x,X)\bar{s}(X))\big). \tag{33}$$

The first line says the mean drift is the average NTK-transmitted source update. The second line says stochastic heat grows when the NTK neighborhood of x sends diverse tangent votes; in the weak-mean regime it reduces to the operator energy

$$\tilde{\mathcal{V}}(x)^2 \approx \eta^2 B\, \mathbb{E}_X\left[s(X)^\top \mathcal{K}(x,X)^\top \mathcal{K}(x,X) s(X)\right]. \tag{34}$$

Radial drift depends on the sign of the transmitted mean vote:

$$\tilde{\mathcal{D}}(x) = \hat{z}_t(x)^\top \mu(x) \approx -\eta B\, \mathbb{E}_X \left[\hat{z}_t(x)^\top \mathcal{K}(x,X)\bar{s}(X)\right]. \tag{35}$$

Move the NTK onto the radial test direction. Since $\bar{s}(X)$ is tangent to the unit sphere at X, only the tangent projection matters:

$$a_K(x,X) := P^{\perp}_{\hat{z}_t(X)} \mathcal{K}(x,X)^\top \hat{z}_t(x), \qquad P^{\perp}_{\hat{z}_t(X)} = I - \hat{z}_t(X)\hat{z}_t(X)^\top. \tag{36}$$

Thus

$$\tilde{\mathcal{D}}(x) \approx \eta B\, \mathbb{E}_X \left[a_K(x,X)^\top(-\bar{s}(X))\right]. \tag{37}$$

Let $\varphi_K(x,X)$ be the angle between $\mathcal{K}(x,X)^\top \hat{z}_t(x)$ and $\hat{z}_t(X)$, and let $\psi_K(x,X)$ be the angle between $a_K(x,X)$ and $-\bar{s}(X)$. Since

$$\|a_K(x,X)\| = \|\mathcal{K}(x,X)^\top \hat{z}_t(x)\| \sin\varphi_K(x,X), \tag{38}$$

the radial drift has the projected full-NTK form

$$\tilde{\mathcal{D}}(x) \approx \eta B\, \mathbb{E}_X \Big[\underbrace{\|\mathcal{K}(x,X)^\top \hat{z}_t(x)\|}_{\text{Transmission}} \ \underbrace{\sin\varphi_K(x,X)}_{\text{Concept direction alignment}} \ \underbrace{\|\bar{s}(X)\|}_{\text{Update size}} \underbrace{\cos\psi_K(x,X)}_{\text{Update alignment}} \Big]. \tag{39}$$

Positive drift needs strong transmission, a large tangent component, a large source update, and positive alignment. Negative alignment compresses the norm; high variance without coherent alignment heats the embedding without creating outward drift.

E Variance Decomposition of the Transmitted Update

Let $\bar{s}(x) := \mathbb{E}[s_k \mid X_k = x]$ and $\Sigma_s(x) := \mathrm{Cov}(s_k \mid X_k = x)$. By the law of total covariance applied to the scale-normalized transmitted vector $\mathcal{K}(x_*, X_k)s_k$:

$$\mathrm{Cov}(\mathcal{K}(x_*, X_k)s_k) = \underbrace{\mathbb{E}_X\left[\mathcal{K}(x_*, X)\Sigma_s(X)\mathcal{K}(x_*, X)^\top\right]}_{\text{intra-batch covariance}} + \underbrace{\mathrm{Cov}_X(\mathcal{K}(x_*, X)\bar{s}(X))}_{\text{index-sampling covariance}}. \tag{40}$$

Taking the trace separates residual batch noise (first term, noise after conditioning on which input is drawn) from index-sampling noise (second term, noise from drawing different inputs with different conditional mean gradients):

$$\begin{aligned}\mathrm{Tr}(\mathrm{Cov}(\mathcal{K}(x_*, X_k)s_k)) &= \mathbb{E}_X\left[\mathrm{Tr}\big(\mathcal{K}(x_*, X)\Sigma_s(X)\mathcal{K}(x_*, X)^\top\big)\right] \\ &\quad + \mathrm{Tr}(\mathrm{Cov}_X(\mathcal{K}(x_*, X)\bar{s}(X))). \end{aligned}\tag{41}$$

The scale-normalized fluctuation in Theorem 1 is B times this trace, up to the fixed batch normalization convention of the loss. The combined second-order task heat also includes the squared mean. With $Y(X) := \mathcal{K}(x_*, X)s(X)$,

$$\mathbb{E}\left[\left\|\sum_{k\in B_t} Y(X_k)\right\|^2\right] \approx B\, \mathbb{E}_X\left[\|Y(X)\|^2\right] + B(B-1)\left\|\mathbb{E}_X[Y(X)]\right\|^2. \tag{42}$$

The first term captures stochastic heating from per-slot fluctuations; the second captures coherent drift amplified by batch size and dominates for hub concepts with large $\|\mathbb{E}[\mathcal{K}s]\|$. The approximation Eq. (15) in the main text keeps the first term in the weak-mean regime.

F Macroscopic Intuition: Why Positive Count Need Not Change Norm

The main equation contains no independent ρ variable: support-size dependence enters only through $\mathcal{D}_{\star,\rho}(x)$ and $\mathcal{V}_{\star,\rho}(x)^2$. Why should those be insensitive to the number of positive contexts?

Intuitively, a concept that appears more often in training provides more gradient samples per unit of training time, which could reduce the per-step variance $\tilde{\mathcal{V}}^2$ of its embedding update—the gradient is averaged over more examples, so its dispersion shrinks. In the heat-dominated regime of Section 4.1, $\tilde{\mathcal{V}}^2 \propto R^2$, so one might expect higher frequency to drive lower norms.

Table 5: Embedding norm is independent of concept frequency across architectures. Norms are reported at each ρ (number of positive contexts) for a fixed concept; values are stable across two orders of magnitude of ρ (max CV $< 2\%$). Training details: mixed-ρ SGD with weight decay, no frozen parameters.

Model	ρ=5	ρ=20	ρ=100	ρ=500	ρ=2000
DistilBERT (base-uncased)	7.346	7.307	7.294	7.303	7.344
E5 (intfloat/e5-base)	2.929	2.934	2.953	2.829	2.893
SigLIP (base-patch16-384)	4.188	4.189	4.191	4.195	4.173
CLIP (ViT-B/32)	23.765	23.768	23.867	23.587	22.946

Table 6: Embedding norms are consistent across languages. Mean-pooled hidden-state norms ($\|\mathbf{z}(x)\|_2$) for 25 parallel sentences in 6 languages. Cross-language CV of language means is below 1% for both models, confirming that norm is driven by semantic content rather than language-specific statistics.

Language	mE5-large-instruct	BGE-M3
English	26.16	25.96
Chinese	26.46	25.67
Spanish	26.14	26.11
French	26.25	26.23
Arabic	26.35	26.11
Hindi	26.47	26.09
Cross-lang CV	0.55%	0.75%
Per-sent CV	1.68%	1.05%

The key quantitative point is that this averaging effect decays rapidly: the empirical measure over a concept's positive contexts converges to its limiting conditional law at rate $O(\rho^{-1/2})$. This means the variance $\tilde{\mathcal{V}}^2_\rho(x)$ is nearly indistinguishable from $\tilde{\mathcal{V}}^2_\infty(x)$ for any ρ above a small threshold. In the synthetic experiments below, ρ ranges from 5 to 2000, yet the per-model coefficient of variation stays under 2%—the frequency signal is simply washed out by the dominant, uniform radial contraction from weight decay.

Thus three conceptually distinct mechanisms govern norm variation: (i) pair count ρ refines the empirical context distribution, converging so fast that it leaves the equilibrium norm unchanged; (ii) semantic breadth changes the limiting conditional law Q_c itself, which *can* shift $\mathcal{D}_\star$ and $\mathcal{V}_\star$ and thereby change R; (iii) frequency can act through sparse-update clock effects driven by the optimizer, but those are optimizer artifacts, not semantic-density laws.

Empirical verification. We train several architectures on synthetic datasets where ρ varies from 5 to 2000 while holding semantic content fixed. Table 5 reports the converged embedding norm at each frequency; norms are essentially flat across two orders of magnitude of ρ.

This consistency also holds across languages for the same semantic content, as demonstrated in Table 6 using multilingual models (multilingual-E5-large-instruct [30] and BGE-M3 [2]) across six languages.

G Optimizer Artifacts: Momentum

Momentum smoothing of stochastic heat. Momentum-style optimizers replace a raw stochastic update U_t by an exponential moving average $m_t = \beta m_{t-1} + (1-\beta)U_t$. Modelling U_t as i.i.d. with mean $\bar{U}$ and covariance Σ_U, the stationary moving average satisfies $\mathbb{E}[m_\infty] = \bar{U}$ and

$$\mathrm{Cov}(m_\infty) = \frac{1-\beta}{1+\beta}\Sigma_U, \tag{43}$$

Derivation. Unrolling the recursion gives $m_\infty = (1-\beta)\sum_{k=0}^{\infty}\beta^k U_{t-k}$, so by independence of the U_t,

$$\mathrm{Cov}(m_\infty) = (1-\beta)^2\sum_{k=0}^{\infty}\beta^{2k}\,\Sigma_U = \frac{(1-\beta)^2}{1-\beta^2}\,\Sigma_U = \frac{1-\beta}{1+\beta}\,\Sigma_U.$$

So at stationarity ($\bar{U}\approx 0$) momentum rescales stochastic heating by $(1-\beta)/(1+\beta)$ without changing which concepts receive larger transmitted variance.

H Experimental Details

Synthetic dataset (Experiments A and B). The synthetic dataset uses artificial concept tokens paired with 100 positive strings. The positives are drawn from diverse template families representing semantically coherent sentence variations (e.g., "researchers compared *pig* and *swine* in animal behaviour studies"). These token types are treated as natural-vocabulary words (not special tokens), added to the tokenizer vocabulary and resized in the embedding table.

Training uses `MultipleNegativesRankingLoss` from `sentence-transformers` [22], SGD with weight decay (SGDW), batch size 256, and learning rate 10^{-3}. Each λ value is trained independently to norm convergence (early stop when $\max_x |\Delta\|z(x)\|| < 5\times10^{-2}$ and relative change $< 1.5\times10^{-2}$ for 5 consecutive checks every 5 epochs, or when the step cap of 700 updates is reached). All experiments use `distilbert-base-uncased` [23] as the base model.

$\tilde{\mathcal{D}}$ and $\tilde{\mathcal{V}}$ measurement. We estimate $\tilde{\mathcal{D}}(x)$ and $\tilde{\mathcal{V}}(x)^2$ via Monte Carlo gradient probing. For n_{batches} random minibatches drawn from the training pairs, we compute the contrastive loss gradient, apply a virtual gradient step of size δ to all parameters, record the full pre-normalisation embedding displacement vector $\Delta z^{(b)} = z'^{(b)} - z$, then restore the parameters. The unit-norm scaled displacement is

$$\Delta\tilde{z}^{(b)} := \frac{\Delta z^{(b)}\cdot\|z\|}{\delta}, \tag{44}$$

which approximates one draw from $\Delta\tilde{z}_{\mathrm{task}}$ in the limit $\delta\to 0$. Let $\bar{\Delta\tilde{z}} = \frac{1}{B}\sum_b \Delta\tilde{z}^{(b)}$ be the sample mean vector. The estimators are

$$\hat{\tilde{\mathcal{D}}}(x) = \hat{z}(x)^\top \bar{\Delta\tilde{z}}, \tag{45}$$

$$\hat{\tilde{\mathcal{V}}}(x)^2 = \frac{1}{B-1}\sum_{b=1}^{B}\|\Delta\tilde{z}^{(b)} - \bar{\Delta\tilde{z}}\|^2. \tag{46}$$

The first line projects the mean displacement onto the radial direction; the second is the trace of the sample covariance of the full d-dimensional displacement, matching the theoretical definition $\tilde{\mathcal{V}}^2 = \mathrm{Tr}(\mathrm{Cov}(\Delta\tilde{z}_{\mathrm{task}}))$ from Theorem 1. Estimating $\tilde{\mathcal{D}}$ from the scalar norm change $\Delta R/(\delta\|z\|)$ recovers $\hat{z}^\top\bar{\Delta\tilde{z}}$ to first order and is equivalent; estimating $\tilde{\mathcal{V}}^2$ from the variance of the scalar $\Delta R/(\delta\|z\|)$ would instead give $\hat{z}^\top\mathrm{Cov}(\Delta\tilde{z})\hat{z}$, the radial component only, and is *not* used here. Because λ_{rad} varies across concept positions and training steps, it is averaged over the converged model by measuring the decay component of the parameter update separately from the task component; details follow the derivation in Eq. (7).

CIFAR-10H uncertainty experiment. We use CLIP [21] (`openai/clip-vit-base-patch32`) to extract image embeddings from the CIFAR-10H [19] test split. Human uncertainty per image is defined as the entropy of the soft label distribution over 10 classes. No fine-tuning or adaptation occurs.

Retrieval experiments. For the LAION text–text comparison, DistilBERT [23] and E5 [28] embeddings are precomputed once and reused for all γ values.

LLM-generated general–specific QA dataset. The controlled specificity-selection experiment uses a dataset of 1,060 query–answer pairs generated by an LLM and curated with an LLM judge. Each item consists of a factual query, a *general* answer, and a *specific* answer. The general answer conveys the essential fact in plain terms; the specific answer adds concrete mechanism, numerical thresholds, named entities, or process detail absent from the general answer. Items are drawn from 17 domains: medicine (150), law (143), computer systems (124), biology (122), chemistry (107), physics (104), machine learning (90), economics (54), mathematics (36), astronomy (30), climate science (27), history (26), public policy (15), statistics (13), engineering (11), geography (5), and finance (3).

Each candidate pair was accepted only when the LLM judge confirmed: (i) both answers are relevant and correct, (ii) the specific answer logically entails the general one but not vice versa, (iii) the specific answer adds concrete information absent from the general answer, (iv) there is no factual contradiction between the two. This acceptance criterion ensures the two answers differ in specificity rather than in relevance or correctness.

To control for the differing text lengths of the two answers (the specific answer tends to be longer), we adopt a *repeated-general* protocol: the general answer is concatenated with itself before encoding, so that both answers have comparable token counts. Mean-pooled models are insensitive to token order, so the repetition leaves both the direction and the norm of the general answer embedding approximately unchanged. For last-token-pooled (decoder) models, the repetition affects the last token, and we report results with this caveat.

For each query–answer pair, we encode the query q, the repeated general answer a_g, and the specific answer a_s with a frozen embedding model. We then compute the paired z-statistic $z = \sqrt{n} \cdot \overline{\Delta}/\sigma_\Delta$ where $\Delta_i = \|a_s^{(i)}\| - \|a_g^{\times 2}\|$, as well as the fraction of pairs where the specific answer has the smaller norm. As a relevance control, we also compute $\cos(q, a_s) - \cos(q, a_g)$; the difference is uniformly near zero across all models, confirming that both answers are equally relevant to the query in direction space.

The experiment is run across 8 embedding models spanning encoder-only (mean-pooled), vision-language, and decoder-only architectures. All embeddings are extracted without fine-tuning.

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